

Preference, Choice and Demand

Advanced Microeconomics 1: Economic Decision Making. Lecture 1

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The questions of this block

1. Who did the 2022 price shock (inflation) hurt, and by how much?
2. Is automatic wage indexation an effective policy for mitigating these impacts?

- These sound like statistics questions but they are not “hurt” is a value judgment that we need to make precise.

Our plan for the next 4 lectures.

1. **Formalise** what we observe: prices, budgets, demand. today
2. **Describe**: whose prices rose, whose budgets are exposed. today
3. **Define the ideal answer**: “as well off as before,” in euros. L2
4. **Connect it to data**: when does observed demand determine that answer? L3–L4
5. **Audit**: if the assumptions are wrong, which way is our answer wrong? Always

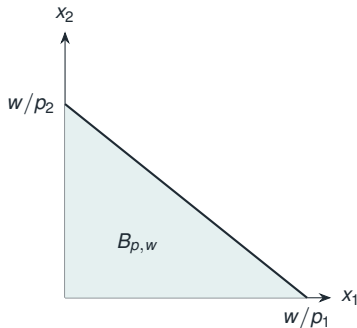
Formalise: demands, prices, wealth

Formalising

- To answer this question (and others like it) we...
 - want to be precise, we need ϵ values.
 - want to be transparent, what exact assumptions do we need (and what happens when they fail).
- To achieve these goals we need to get formal!

- L commodities, described by a consumption set $X = \mathbb{R}_+^L$.
- L prices $p \in \mathbb{R}_{++}^L$.
- **Walrasian budget set**, $p \gg 0$, $w > 0$:

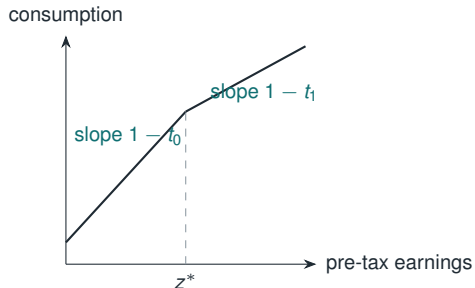
$$B_{p,w} = \{x \in \mathbb{R}_+^L : p \cdot x \leq w\}.$$



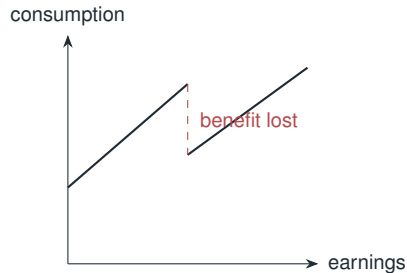
Real budget sets are not hyperplanes (straight)

formalise

Kink (e.g. progressive tax)



Notch (benefit cliff)



Walrasian demand and two accounting laws

Definition · Walrasian (Marshallian) demand

$x(p, w) \subseteq B_{p,w}$: the chosen bundle(s) at each price–wealth pair.

Two assumptions:

1. **Homogeneity of degree zero:** $x(\alpha p, \alpha w) = x(p, w)$ — only relative prices and real wealth matter; “no money illusion.”
2. **Walras’ law:** $p \cdot x(p, w) = w \rightarrow$ you spend everything.

The four formal objects

- Consumption set: $X \in \mathbb{R}_+^L$.
- Prices: $p \in \mathbb{R}_{++}^L$.
- Budget set: $B_{p,w} = \{x \in X : p \cdot x \leq w\}$.
- Demand: $x(p, w) \subseteq B_{p,w}$.

What can we already learn: commodities

Describing commodities

Comparative statics: how does behaviour change when the economic parameters change?

How x_l varies with p_k and w tells us about the properties of commodity l .

Changing wealth — some definitions

Keep p constant at $\bar{p}$, vary w , then ...

- $x_I(\bar{p}, w)$ is the Engel curve for commodity I
- $E_{\bar{p}} = \{x(\bar{p}, w) : w > 0\}$ is the wealth expansion path
- $\frac{\partial x_I(p, w)}{\partial w}$ is the wealth effect for commodity I
- $\epsilon_{Iw}(p, w) = \frac{\partial x_I(p, w)}{\partial w} \frac{w}{x_I(p, w)}$ is the elasticity of demand w.r.t. wealth.
- $b_I(p, w) = \frac{p_I x_I(p, w)}{w}$ is the budget share of I .

Changing wealth — categorising commodities

A commodity l is a ...

- normal commodity if $\frac{\partial x_l(p, w)}{\partial w} \geq 0$
- inferior commodity if $\frac{\partial x_l(p, w)}{\partial w} < 0$
- luxury commodity if $\epsilon_{lw}(p, w) \geq 1$, budget share increases in wealth
- necessity commodity if $\epsilon_{lw}(p, w) < 1$, budget share decreases in wealth

Engel (1857): food's share falls with wealth — robust across countries and centuries ($\sim 50\%$ in poor countries, $\sim 15\%$ in rich ones). Within Belgian quartiles, home energy does the falling ($10\% \rightarrow 6\%$) and restaurants the rising ($5\% \rightarrow 10\%$).

Changing prices — some definitions

- $\frac{\delta x_l(p, w)}{\delta p_k}$ is the price effect of p_k on commodity l .
- $\epsilon_{lk}(p, w) = \frac{\delta x_l(p, w)}{\delta p_k} \frac{p_k}{x_l(p, w)}$ is the elasticity of demand for l with respect to p_k .
 - ϵ_{ll} = own price elasticity
 - ϵ_{lk} = cross price elasticity

Changing prices — categorising commodities

- Demand for l is elastic if $\epsilon_{ll}(p, w) \leq -1$.
- Demand for l is inelastic if $-1 < \epsilon_{ll}(p, w) \leq 0$.
- Demand for l is a Giffen good if $\epsilon_{ll}(p, w) > 0$.

Home energy in the short run: about as inelastic as demand gets — you cannot quickly stop heating. Inelastic + large budget share $\Rightarrow$ a price shock is going to hurt.

First payoff: your own personal inflation rate

Let household h have budget shares b_l^h . For small price changes, the inflation *that household faces* is

$$\pi^h = \sum_{l=1}^L b_l^h \pi_l \quad (\text{share-weighted price growth}).$$

- Different $b^h \Rightarrow$ different π^h .
- The official index does this with *aggregate expenditure* weights — so each euro, not each household, gets one vote. A rich household's basket counts more.

The load-bearing assumption

Why we now need an assumption

“Worse off” — by whose standard?

- The compensation question is about *welfare*: each household's own ranking of situations. We can't observe this, we only see the choices you make.
- Everything rests on these choices giving us useful information. One key assumption **rationality**.

Lets get formal

What, exactly, do we mean by you prefer X ? What are preferences?

Preferences

Definition · preference relation

$\succsim$ is a binary relation on X ; $x \succsim y$ reads “ x is at least as good as y .”

- x is better than y : $x \succ y \Leftrightarrow x \succsim y$ and not $y \succsim x$.
- x is indifferent to y : $x \sim y \Leftrightarrow x \succsim y$ and $y \succsim x$.

To answer our question, we need each household h 's own ranking, $\succsim_h$.

Rationality

Definition · rationality

The preference relation, $\succsim$, is *rational* if $\forall x, y, z \in X$ it is:

complete: $x \succsim y$ or $y \succsim x$

transitive: $x \succsim y \succsim z \Rightarrow x \succsim z$

Seems reasonable?

Definition · utility representation

$u : X \rightarrow \mathbb{R}$ represents $\succsim$ if $x \succsim y \iff u(x) \geq u(y)$.

Why do we want a u ? We need numbers, euros!

Utility representation

Proposition · MWG 1.B.2

If a utility function $u : X \rightarrow \mathbb{R}$ represents $\succsim$, then $\succsim$ is rational.

Proof $\rightarrow$ p. 35

$\succsim$ inherits the properties of $\geq$ on $\mathbb{R}$ through the representation.

What does rationality look like

What would rationality look like in the data?

Our data: pairs (p^t, x^t) — prices faced, bundle chosen — or an estimated demand function $x(p, w)$.

- Choosing x when y was **affordable** reveals $x \succsim y$. This is the information we get!
- Rationality $\Rightarrow$ no contradictions.

The weak axiom

Definition · WARP for demand functions (MWG 2.F.1)

$x(p, w)$ satisfies WARP if for all pairs (p, w) and (p', w') we have that if:

$$p \cdot x(p', w') \leq w \text{ and } x(p', w') \neq x(p, w)$$

Then

$$p' \cdot x(p, w) > w'.$$

If the bundle chosen at (p', w') was also affordable at (p, w) but was not chosen at (p, w) then $x(p, w)$ is revealed preferred to $x(p', w')$. Then it cannot be that $x(p', w')$ is also revealed preferred to $x(p, w)$ that is when $x(p', w')$ was chosen it must have been that $x(p, w)$ was not affordable i.e. $p' \cdot x(p, w) > w'$.

The compensation problem, first pass

Compensated price changes

A price change does two things: changes relative prices, and changes real wealth.

Definition · (Slutsky) compensated price change

From (p, w) with $x = x(p, w)$, move to p' and set $w' = p' \cdot x$: wealth adjusted so *the old bundle remains exactly affordable*. The adjustment $\Delta w = \Delta p \cdot x(p, w)$ is the Slutsky wealth compensation.

- Pay each household enough that its old basket stays affordable at the new prices. Automatic indexation is a country-sized Slutsky compensation.

What does a compensated household do

Theorem · MWG 2.F.1

Suppose $x(p, w)$ is h.d.0 and satisfies Walras' law. Then x satisfies WARP $\iff$ for every compensated price change,

$$(p' - p) \cdot [x(p', w') - x(p, w)] \leq 0, \quad \text{strict if } x(p', w') \neq x(p, w).$$

Proof $\rightarrow$ p. 36

- A **law of demand**: a compensated household tilts away from what got relatively more expensive ($\Delta p_\ell > 0 \Rightarrow \Delta x_\ell \leq 0$).
- So if you hand a household its Slutsky compensation it will **not** rebuy the old basket — it substitutes.
- A compensated household will buy a new x , and have a new utility.

Basket indexation over-compensates

Compensate a household so its old bundle x stays affordable ($w' = p' \cdot x$). If it chooses $x' \neq x$, then it chose x' *while x was affordable*: by revealed preference, the household is at least as well off as before. The true cost of staying as-well-off is lower, because people substitute.

- So automatic indexation is too generous.
- How much over-compensation?

Measuring substitution: the compensated derivative

The over-payment above is exactly what a fixed basket ignores: *substitution*. To quantify it we need the derivative of demand **along a compensated change**.

1. **Why compensated?** The only law we have proved — 2.F.1 — holds along *compensated* changes. If we want to use it, we must move along one.
2. **What the compensation is.** Keep the old bundle x just affordable. At prices $p + dp$ it costs $(p + dp) \cdot x = w + dp \cdot x$, so wealth must move by exactly

$$dw = dp \cdot x.$$

3. **How demand responds.** Prices *and* wealth both moved, so to first order

$$dx = \underbrace{D_p x dp}_{\text{prices moved}} + \underbrace{D_w x dw}_{\text{wealth moved}} = [D_p x + D_w x x^T] dp,$$

substituting $dw = dp \cdot x$ from step 2 into the second term.

The Slutsky matrix

Definition · Slutsky substitution matrix

$$S(p, w) = D_p x(p, w) + D_w x(p, w) x(p, w)^T, \quad s_{\ell k} = \frac{\partial x_\ell}{\partial p_k} + \frac{\partial x_\ell}{\partial w} x_k.$$

- Step 3 now reads $dx = S(p, w) dp$:
- Feed that into the law. 2.F.1 gives $dp \cdot dx \leq 0$; substituting $dx = S dp$,

$$dp^T S(p, w) dp \leq 0 \quad \text{for every price step } dp.$$

- S quantifies substitution; in Lecture 3 we will use it to correct the naive index.

What the Slutsky matrix can tell us

Proposition · MWG 2.F.2–2.F.3

If x is h.d.0, satisfies Walras' law and WARP, then $S(p, w)$ is **negative semidefinite** ($v \cdot Sv \leq 0$) and **singular**: $S p = 0$, $p^T S = 0$.

Proof → p.37

So if demand x satisfies *h.d.0*, Walras, WARP then...

- NSD $\Rightarrow$ naive index (weakly) over compensates any price change.
- Singular $\Rightarrow$ only relative price changes cause issues.

Summary

Takeaways

- **We have formalised the problem.**
- **Categorised commodities.**
- Indexation will overcompensate due to substitution by an amount dictated by S
- Next: Quantify.

Appendix: proofs

Proof: utility representation $\Rightarrow$ rationality (MWG 1.B.2)

Recall: u represents $\succsim$ means $x \succsim y \iff u(x) \geq u(y)$.

- *Completeness.* Take any $x, y \in X$. Then $u(x)$ and $u(y)$ are real numbers, and $\geq$ on $\mathbb{R}$ is complete: $u(x) \geq u(y)$ or $u(y) \geq u(x)$. Reading the representation right-to-left, $x \succsim y$ or $y \succsim x$.
- *Transitivity.* Suppose $x \succsim y$ and $y \succsim z$. By the representation, $u(x) \geq u(y)$ and $u(y) \geq u(z)$; transitivity of $\geq$ on $\mathbb{R}$ gives $u(x) \geq u(z)$; right-to-left again, $x \succsim z$. ■

Proof: WARP $\Rightarrow$ compensated law (MWG 2.F.1, direction (i))

If $x(p', w') = x(p, w)$ the claim is trivial. So take $x' \equiv x(p', w') \neq x \equiv x(p, w)$ and expand:

$$(p' - p) \cdot (x' - x) = p' \cdot (x' - x) - p \cdot (x' - x).$$

- **First term is zero:** $p' \cdot x' = w'$ (Walras) and $p' \cdot x = w'$ (definition of the compensation). So the old bundle is *exactly affordable* at the new situation.
- **Second term is positive:** x affordable at (p', w') but x' chosen there $\Rightarrow$ (WARP) x' was *not* affordable at (p, w) : $p \cdot x' > w = p \cdot x$. Hence $p \cdot (x' - x) > 0$.
- Together: $(p' - p) \cdot (x' - x) = 0 - (\text{positive}) < 0$. ■

Direction (ii), compensated law $\Rightarrow$ WARP, is tedious and not extremely insightful; see MWG.

Proof: WARP $\Rightarrow$ S negative semidefinite (MWG 2.F.2)

Fix (p, w) , take any direction $v \in \mathbb{R}^L$ and a small price step $dp = \varepsilon v$, *compensated*:
 $dw = x(p, w) \cdot dp$.

- Demand is differentiable, so the induced change in demand is, to first order,

$$dx = D_p x dp + D_w x dw = [D_p x + D_w x x^T] dp = S(p, w) dp.$$

- The change is a compensated price change, so the compensated law of demand (Theorem 2.F.1, which used WARP + h.d.0 + Walras) applies:

$$dp \cdot dx \leq 0 \quad \Rightarrow \quad \varepsilon^2 (v \cdot S(p, w) v) \leq 0.$$

- $\varepsilon^2 > 0$ and v was arbitrary: $v \cdot S v \leq 0$ for all v . ■

Proof: $Sp = 0$ and $p^T S = 0$ (from h.d.0 and Walras; MWG 2.E)

Neither part uses WARP — pure accounting.

- **h.d.0:** $x(\alpha p, \alpha w) = x(p, w)$ for all $\alpha > 0$. Differentiate in α at $\alpha = 1$:

$$D_p x p + D_w x w = 0. \quad (\text{E1})$$

- **Walras:** $p \cdot x(p, w) = w$ identically. Differentiate in w , and in p :

$$p^T D_w x = 1, \quad p^T D_p x + x^T = 0. \quad (\text{E2}, \text{E3})$$

- **Right-multiply by p :** $Sp = D_p x p + D_w x (x \cdot p) \stackrel{\text{Walras}}{=} D_p x p + D_w x w \stackrel{(\text{E1})}{=} 0$.

- **Left-multiply by p^T :** $p^T S = p^T D_p x + (p^T D_w x) x^T \stackrel{(\text{E2})}{=} p^T D_p x + x^T \stackrel{(\text{E3})}{=} 0$. ■